

Session 4 (Live demo): the 2×2 calculator

Foundations of Clinical Research · Week 4 · Saturday · Neudata · Instructor script

Slots	Main demo (8 min): section 4.2, just before the break, slide "Live demo: the 2x2 calculator" (sheet <i>Association</i>). Short add-on (1–2 min, optional): section 4.3, slide "Same test, different setting: PPV depends on prevalence" (sheet <i>Diagnostic</i>).
File	Course/Resources/tools/2x2_calculator.xlsx (regenerate with <code>python make_2x2_calculator.py</code> in the same folder)
Software	Excel or LibreOffice Calc (any recent version). No internet needed.
Participants	Watch only. No laptops needed.
Goal	Everyone sees risk, RR, RD, NNT and OR computed from ONE table and hears each one in plain words. In 4.3 they see the same test give very different PPVs in two settings.

Before the session (5 minutes, the day before)

1. Open the file and check that it shows the **worked example**: Restrictive 20 / 80, Usual care 30 / 70 (sheet **Association**); 18 / 16 / 2 / 64 and a new-setting prevalence of **2%** (sheet **Diagnostic**). If someone has changed it, rerun `make_2x2_calculator.py`.
2. Set the zoom to about **150%** so the back row can read it. Close every other window.
3. Only the yellow cells can be edited. The sheets are protected without a password (Review → Unprotect Sheet if you need to change a label).
4. Keep a whiteboard marker ready for the fallback.

Part 1: Main demo in 4.2 (8 minutes, sheet *Association*)

Step 1: Show the layout (1 minute)

Do: open sheet **Association**. Point at the yellow cells.

Say: "Same layout as the slide: treatment or exposure in rows, with the treated group on top, and outcome in columns, with 'died' on the left. a, b, c and d are always in the same place."

Appears: the 2×2 table with 20 | 80 and 30 | 70, totals 100 and 100.

Step 2: Read the results in words (3 minutes)

Do: point at each row of the results table, reading the right-hand "In plain words" column aloud.

Row	Value shown	Plain words shown	Teaching point
Risk in exposed / treated	20.0%	20 of 100 patients in the Restrictive group had the outcome	Risk is out of everyone in the group.

Row	Value shown	Plain words shown	Teaching point
Risk in unexposed / control	30.0%	30 of 100 ... Usual care ...	
Risk ratio (RR)	0.67	The risk is 33% LOWER in the exposed/treated group	Relative effect.
Risk difference (RD)	-10.0%	10.0 fewer patients with the outcome per 100	Absolute effect: the one patients need.
NNT / NNH	10.0	NNT: treat about 10 patients to PREVENT one outcome	
Odds (two rows)	0.25 and 0.43	Odds compare those WITH the outcome to those WITHOUT	Odds are not risks.
Odds ratio (OR)	0.58	outcome is common, so OR lies further from 1 than RR	Matches the OR-vs-RR chart.

Step 3: Make the outcome rare (2 minutes)

Do: type **2** in a (Restrictive, died), **98** in b, **3** in c (Usual care, died), **97** in d. Press Enter after each.

Appears:

	Before	After
Risks	20% vs 30%	2% vs 3%
RR	0.67	0.67 (unchanged)
RD	-10%	-1%
NNT	10	100
OR	0.58	0.66, now close to RR

Say: "Same relative effect, but now we must treat 100 patients to prevent one death. And because death is now rare, the OR has almost caught up with the RR."

Do: type the original numbers back (20, 80, 30, 70).

Step 4: Case-control warning (1 minute)

Do: point at the red line under the table.

Say: "The spreadsheet will happily calculate a 'risk' from case-control data. Do not read it, because the researchers chose how many cases and controls to include. In your second calculation, only the OR is valid. After the break you will do two calculations yourselves (Part C of your handout)."

Part 2: Add-on in 4.3 (1–2 minutes, optional, sheet *Diagnostic*)

Use this only if the laptop is already open and you are on time. The slide makes the same point without it.

1. **Do:** click the **Diagnostic** tab. **Appears:** sensitivity 90.0%, specificity 80.0%, PPV 52.9%, NPV 97.0%, prevalence 20.0% (the ICU picture).
2. **Do:** point at the blue box *"Same test, another setting"*. The yellow prevalence cell shows **2%**. **Appears:** PPV in the new setting **8.4%**, NPV **99.7%**.
3. **Say:** *"Same sensitivity, same specificity. Screening a population where 2 in 100 have the disease, fewer than 1 in 10 positive results are real."*
4. **Do (optional):** point at the small table underneath: prevalence 1% / 2% / 5% / 20% / 50% gives PPV 4.3% / 8.4% / 19.1% / 52.9% / 81.8%.

Teaching points to land

1. **One layout, every measure.** Most errors are layout errors, not arithmetic errors.
2. **Relative versus absolute.** An RR of 0.67 can mean an NNT of 10 or of 100, depending on the baseline risk.
3. **OR $\approx$ RR only when the outcome is rare.**
4. **Case-control data: OR only.**
5. **Sensitivity and specificity belong to the test. PPV and NPV depend on how common the disease is.**

After the calculation exercise: checking the answers (about 8 minutes, after the break)

Type each calculation into the Association sheet in front of the room:

Exercise (handout Part C)	Type	Shows
Calculation 1 (mattress trial)	a 10, b 190, c 20, d 180	RR 0.50, RD -5.0%, NNT 20, OR 0.47
Calculation 2 (candidaemia case-control)	a 24, b 16, c 16, d 64	OR 6.00 (ignore the "risks")

Discuss each result in words (one sentence each), including why Calculation 2 gives only an OR. Then restore the worked example (20, 80, 30, 70) so the file is ready for the next course.

Fallback: no projector or no spreadsheet

Draw this on the whiteboard and work through it with the room. The numbers are chosen to be easy:

	Died	Survived	Total	
Restrictive	20	80	100	risk 20/100 = 0.20
Usual care	30	70	100	risk 30/100 = 0.30
RR = 0.20 / 0.30 = 0.67				
RD = 0.20 - 0.30 = -0.10 (10 fewer deaths per 100)				
NNT = 1 / 0.10 = 10				
OR = (20 x 70) / (80 x 30) = 1400 / 2400 = 0.58				

For the rare-outcome point, write 2/100 vs 3/100: RR = 0.67, NNT = 100, OR = (2 x 97) / (98 x 3) = 194 / 294 = 0.66.

For the PPV point, use the slide "Same test, different setting" (PPV $90/170 = 53\%$ in the ICU; $9/107 = 8\%$ in screening).

Common problems

Problem	Fix
Cells show –	One of the four counts is empty or zero in a denominator. Type a number in every yellow cell.
Cannot type in a cell	Only yellow cells are editable (protected sheet). Review → Unprotect Sheet (no password).
Formulas show as text (LibreOffice)	Tools → Options → LibreOffice Calc → View: untick "Formulas"; or press Ctrl + `.
Prevalence typed as 2 instead of 2%	Type 2% or 0.02. The cell is formatted as a percentage.
Numbers look wrong	Check the layout: treated/exposed on top, "died" on the left.